COMS 4776 Lecture 14: Adapting Transformers

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Adapting transformers: Methods and objectives

Adaptation methods:

- **Open** models:
  - Supervised fine-tuning
  - Low-rank adaptation
- **Closed** models:
  - Prompt engineering
  - Retrieval-augmented generaton
  - In-context learning
  - Soft-prompting
  - Distillation

Adaptation objectives

Supervised fine-tuning

Simple idea:

- pre-trained base model
- continuing training on a targeted dataset
- standard objective (next-token prediction, cross-entropy loss)

Issues:

- trade-off between specialization and generalization
- model found to "forget" how to do tasks not in SFT data

Mitigation strategies?

Supervised fine-tuning

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Mitigation strategies:

- careful dataset curation, including replay
- regularization

Low Rank Adapters

- Low-Rank Adapters (LoRA): way to fine-tune transformer without updating all parameters
- alternative to updating $d \times d$ weight matrix W (e.g., attention layers) during fine-tuning
- instead freeze W and add a low-rank update to it:

$$W' = W + AB$$

  - A: small $(d \times r)$ trainable matrix
  - B: small $(r \times d)$ trainable matrix
  - rank r much less than d

Low Rank Adapters: Applications

- Low-Rank Adapters (LoRA) used in LLMs without full fine-tuning
- used for fine-tuning to particular domains
- also used for multi-task learning, with separate LoRA modules per task
- Resource-efficient: only need to store and load LoRA weights

Open versus Closed Models

Closed models have rapidly become the norm (GPT, Claude, Gemini)

- Competitive advantage
- IP, trade secrets
- Financial outlay, income
- Liability

But:

- Open science, research (vs. power concentration)
- Trust, transparency
- Allow customization & innovation

Open models: Llama, Mistral, Qwen

Adapting closed models: Prompt engineering

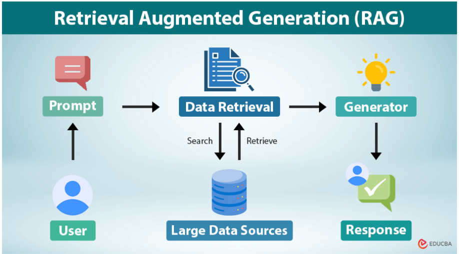
Trial-and-error experimenting with inputs/context/prompts

- Since release of ChatGPT (Nov 2022), dominant mode of adapting the model to own use
- Aimed to shape outputs: correctness, safety, style, etc.

Wide variety, including:

- instructions; background/context; output format; personalization
- constraints and guardrails
- iterative refinement: supply decomposition of problem into steps
- Chain of Thought: "Solve this step-by-step, showing your work"

Retrieval Augmented Generation (RAG)



- Can access databases, web searches, images
- How to find relevant documents (keyword, semantic similarity)
- How to break documents into chunks – incorporate into generation

Advanced RAG:

- Query engineering
- Rerank multiple retrievals

Utility?

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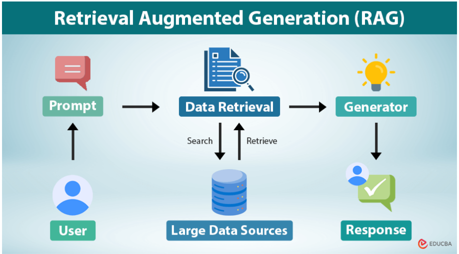

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Advanced RAG:

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Utility: private data, knowledge cut-off date, hallucinations

In-Context Learning (ICL)

Definition: model performs new tasks from examples or instructions in the input prompt,without any parameter updates or fine-tuning Example:

Translate to French

- Hello $\rightarrow$ Bonjour
- Goodbye $\rightarrow$ Au revoir
- Thank You $\rightarrow$ Merci
- Good morning $\rightarrow$

Summary: few-shot form of prompt-engineering; inference-time adaptation

Soft Prompting

Add trainable prompts to the input of a language model

- Learnable continuous vectors prepended to input embeddings
- Live in the same embedding space as tokens
- Not tied to specific words
- Optimized through training

Advantages: parameter efficient, modular

Other terms: prompt tuning; learnable prompts; soft tokens

Related concept is Prefix tuning: add soft prompts at every layer's attention

Model Compression

- Original paper: *Model Compression* by Bucila et al, 2006.
- Best performing model is often a massive ensemble of weak learners
- Aim to develop single model with considerably fewer parameters: reduced space and time complexity at run time
- Focus on generating synthetic data that resembles real data
- Use large model to label new data
- Contains key idea: train small neural net that mimics large complex model

Distillation

- *Distilling the Knowledge of a Neural Network* by Hinton et al, 2015
- Consider softmax outputs of neural net classifier, with temperature T:

$$q_i = \frac{\exp(z_i/T)}{\sum_k \exp(z_k/T)}$$

- higher values of T produces softer distribution across classes
- large model has logits v_i that produce soft probabilities p_i
- cross-entropy gradients of each class logit z_i of distilled model:

$$\frac{\partial C}{\partial z_i} = \frac{1}{T}(q_i - p_i) = \frac{1}{T}\left(\frac{\exp(z_i/T)}{\sum_k \exp(z_k/T)} - \frac{\exp(v_i/T)}{\sum_k \exp(v_k/T)}\right)$$

Distillation

- if T is high relative to logit magnitudes

$$\begin{aligned}
\frac{\partial C}{\partial z_i} &\approx \frac{1}{T}\left(\frac{1 + z_i/T}{N + \sum_k z_k/T} - \frac{1 + v_i/T}{N + \sum_k v_k/T}\right) \\
&\approx \frac{1}{NT^2}(z_i - v_i)
\end{aligned}$$

- assuming logits for each case have zero mean
- so in high T, equivalent to minimizing logit differences
- training objective typically combines soft targets (large model outputs) and hard targets (original labels)

Distillation: Applications

- can use much less data to train distilled model

System & training set	Train Frame Accuracy	Test Frame Accuracy
Baseline (100% of training set)	63.4%	58.9%
Baseline (3% of training set)	67.3%	44.5%
Soft Targets (3% of training set)	65.4%	57.0%

- possibly can also train specialized model on small dataset by training it with soft targets for the non-special classes
- can transfer advanced capabilities of leading proprietary models to smaller, more accessible open source models, or create efficient deployable models
- other applications focus on distilling intermediate representations

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Adaptation objectives

Preference Optimization: RLHF

Alternative to next token prediction, takes on various forms

Reinforcement Learning with Human Feedback (RLHF)

- Human labelers rank various model responses to given prompt
- Train *reward model* to predict which outputs humans will prefer
- Update base model to generate outputs that get higher rewards
- *Alignment* with human preferences

$$\max_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi_\theta(y|x)} \left[r_\phi(x, y) \right] - \beta \mathbb{D}_{\text{KL}} \left[\pi_\theta(y \mid x) \mid\mid \pi_{\text{ref}}(y \mid x) \right]$$

- Maximize expected reward of good responses
- While keeping model responses close to base model
- Train using RL method, such as Proximal Policy Optimization

Preference Optimization: DPO

Direct Preference Optimization (DPO):

- Skip training reward model
- Classification model approach: cross-entropy loss to favor good over bad responses

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

- *u* is upper preferred response, *l* is lower
- π_θ is probability model assigns based on parameters
- compared to reference or based model
- β controls how much the model can differ from the reference model

DPO is efficient, more stable than RLHF

Example of Adapting an LLM

Opinion QA dataset

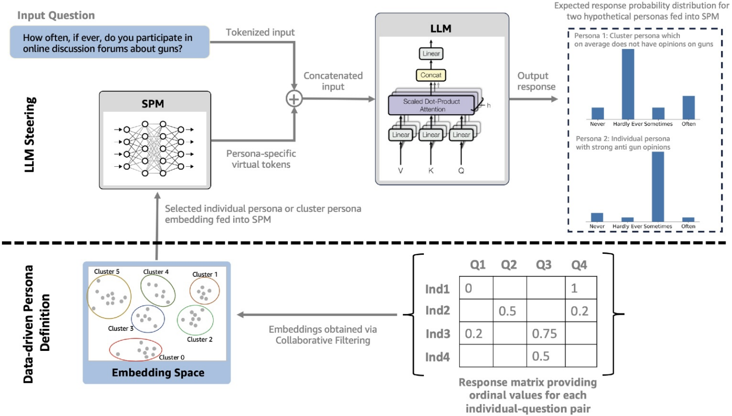
(18K participants, 1500 questions; includes demographic info)

Question	Overall Population Response
How confident are you, if at all, that the actions taken by the international community will significantly reduce the effects of global climate change?	Very confident: 25.32% **Somewhat confident: 41.1%** Not too confident: 31.73% Not at all confident: 1.83%
In general, how often, if ever, would you say you have parties or get-togethers with any of your neighbors?	Every day: 12.33% Several times a week: 11.1% About once a week: 17.87% About once a month: 17.94% **Less than once a month: 28.73%** Never: 12.0%
How enthusiastic are you, if at all, about the possibility of using computer programs to make hiring decisions for society as a whole?	Very enthusiastic: 27.55% **Somewhat enthusiastic: 33.07%** Not too enthusiastic: 38.58% Not at all enthusiastic: 0.78%

LLMs gain implicit average viewpoint:

e.g., under-represents individuals over 65, Mormons, widowed (Santurkar et al, 2023)

Steering LLMs with Personality



On the steerability of large language models toward data-driven personas

Junyi Li, Ninareh Mehrabi, Charith Peris, Palash Goyal, Kai-Wei Chang, Aram Galstyan, Richard Zemel, Rahul Gupta

EMNLP, 2024.